

Math 2551 Worksheet: Linearization and Tangent Planes

1. Find the linearization of $f(x, y) = e^{2y-x}$ at $(1, 2)$.
2. Find the linearization of $f(x, y, z) = \tan^{-1}(xyz)$ at $(1, 1, 0)$.
3. Use the linearization to approximate $f(2.95, 7.1)$ for the function $f(x, y) = \sqrt{x^2 + y}$, knowing that $f(3, 7) = 4$.
4. Graph the function $z = x^{1/3}y^{1/3}$. Examine the graph at $(0, 0)$ - does it look like the function can be well-approximated by a tangent plane there? Based on your conclusion, is this function differentiable at $(0, 0)$?
5. Find the equation for the plane tangent to the graph of $f(x, y) = \sqrt{y - x}$ when $x = 1$ and $y = 2$.